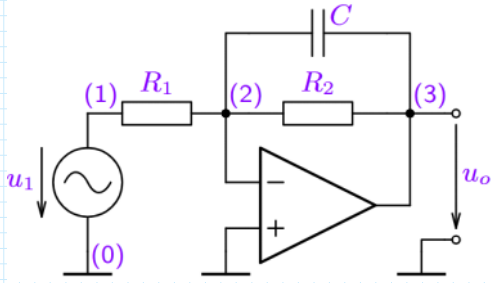


AC_OZ1

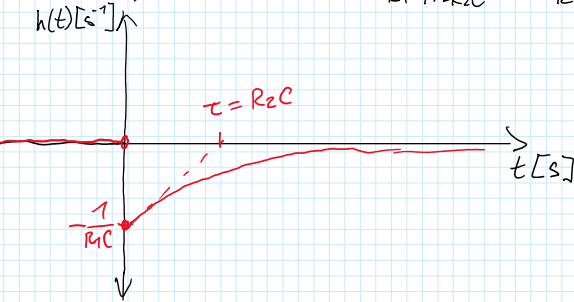


$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} -\frac{R_2}{R_1} \frac{s}{1+sR_2C} = -\frac{1}{R_2C}$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} -\frac{R_2}{R_1} \frac{s}{1+sR_2C} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} -\frac{R_2}{R_1} \frac{1}{1+sR_2C} = 0$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} -\frac{R_2}{R_1} \frac{1}{1+sR_2C} = -\frac{R_2}{R_1}$$



$$H(s) = -\frac{\frac{R_2}{R_1} \frac{1}{sR_2C}}{1 + \frac{1}{sR_2C}} = -\frac{R_2}{R_1} \frac{1}{1+sR_2C} = -\frac{1}{R_1C} \frac{1}{\frac{1}{R_2C} + s}$$

$$h(t) = -\frac{1}{R_1C} e^{-\frac{t}{R_2C}} \mathbb{1}(t)$$

$$W(s) = -\frac{R_2}{R_1} \frac{1}{s} \frac{1}{1+sR_2C} = \frac{A}{s} + \frac{B}{1+sR_2C}$$

$$= -\frac{R_2}{R_1} \frac{1}{s} + \frac{R_2^2C}{R_1} \frac{1}{1+sR_2C} = -\frac{R_2}{R_1} \frac{1}{s} + \frac{R_2}{R_1} \frac{1}{\frac{1}{R_2C} + s}$$

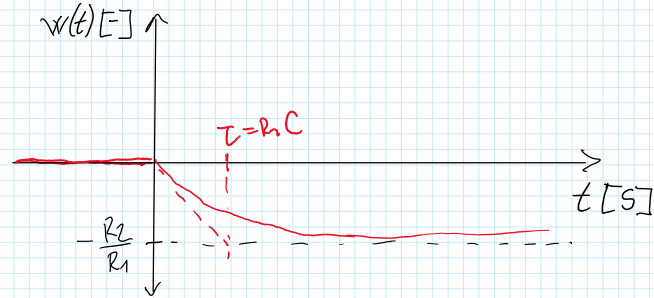
$$w(t) = -\frac{R_2}{R_1} \mathbb{1}(t) + \frac{R_2}{R_1} e^{-\frac{t}{R_2C}} \mathbb{1}(t) = \frac{R_2}{R_1} \mathbb{1}(t) (1 - e^{-\frac{t}{R_2C}})$$

$$-\frac{R_2}{R_1} = A(1+sR_2C) + Bs$$

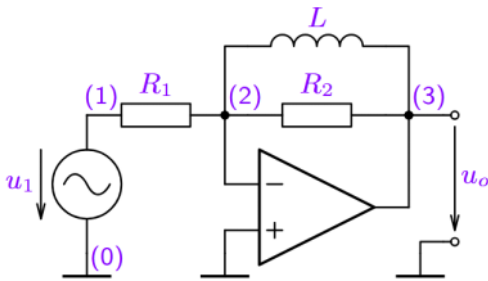
$$A = -\frac{R_2}{R_1}$$

$$AR_2C + B = 0$$

$$B = +R_2C \frac{R_2}{R_1}$$



AC_OZ2

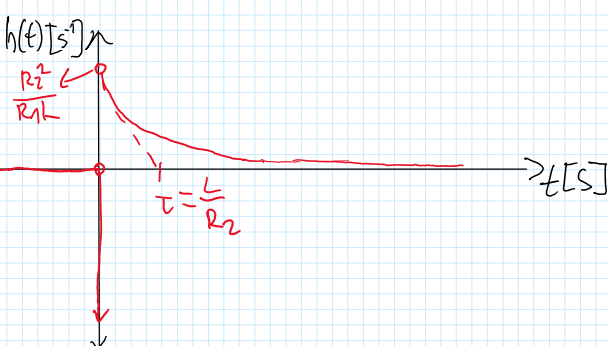


$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} -\frac{s^2 R_2 L}{R_1 R_2 + s R_1 L} = -\infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} -\frac{s^2 R_2 L}{R_1 R_2 + s R_1 L} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} -\frac{s R_2 L}{R_1 R_2 + s R_1 L} = -\frac{R_2}{R_1}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} -\frac{s R_2 L}{R_1 R_2 + s R_1 L} = 0$$



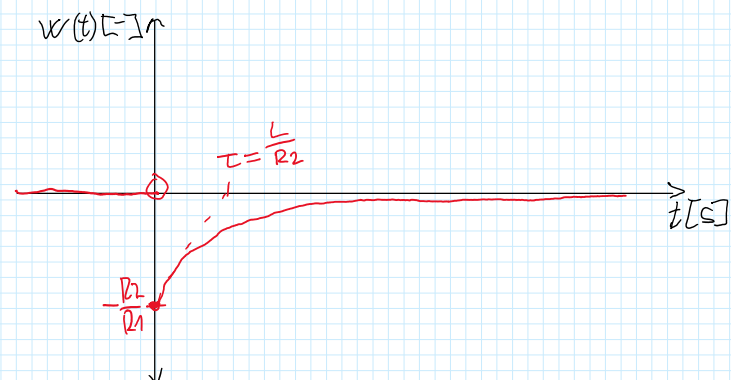
$$H(s) = -\frac{\frac{s R_2 L}{R_2 + s L}}{R_1} = -\frac{s R_2 L}{R_1 R_2 + s R_1 L} = -\frac{R_2}{R_1} + \frac{R_2^2}{R_1 R_2 + s R_1 L}$$

$$= -\frac{R_2}{R_1} + \frac{R_2^2}{R_1 L} \frac{1}{\frac{R_2}{L} + s}$$

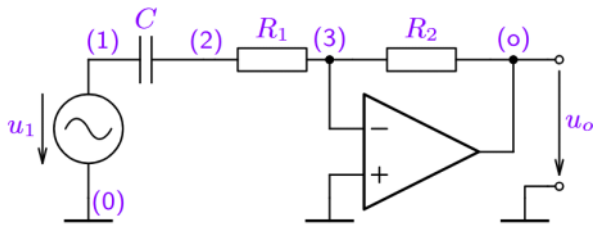
$$h(t) = -\frac{R_2}{R_1} \delta(t) + \frac{R_2^2}{R_1 L} e^{-\frac{R_2}{L} t} \mathbb{1}(t)$$

$$W(s) = -\frac{R_2 L}{R_1 R_2 + s R_1 L} = -\frac{R_2/R_1}{\frac{R_2}{L} + s}$$

$$w(t) = -\frac{R_2}{R_1} e^{-\frac{R_2}{L} t} \mathbb{1}(t)$$



AC_OZ3



$$H(s) = \frac{-R_2}{R_1 + \frac{1}{sC}} = \frac{-sR_2C}{1 + sR_1C} = -\frac{R_2}{R_1} + \frac{R_2}{R_1} \frac{1}{1 + sR_1C}$$

$$h(t) = -\frac{R_2}{R_1} \delta(t) + \frac{R_2}{R_1^2 C} e^{-\frac{t}{R_1 C}} \mathbb{1}(t)$$

$$W(s) = \frac{-R_2 C}{1 + sR_1 C}$$

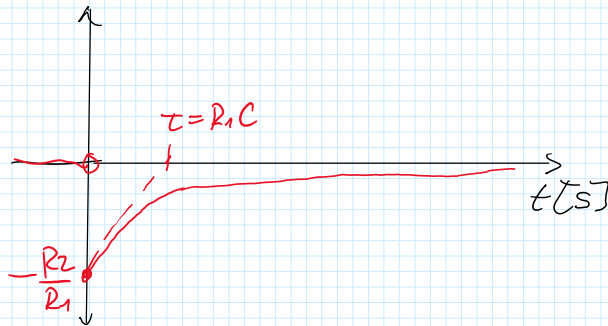
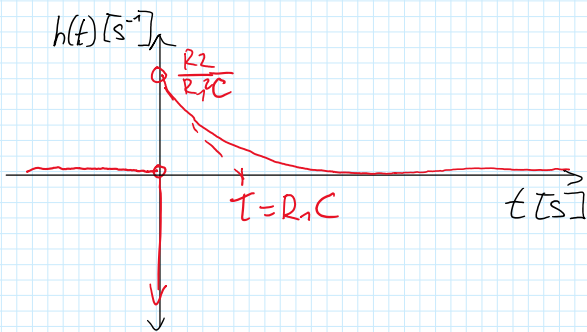
$$w(t) = -\frac{R_2}{R_1} e^{-\frac{t}{R_1 C}} \mathbb{1}(t)$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} -\frac{sR_2C}{1 + sR_1C} = -\infty$$

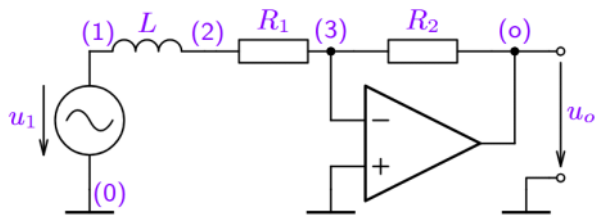
$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} -\frac{s^2 R_2 C}{1 + sR_1 C} = 0$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} -\frac{sR_2C}{1 + sR_1C} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} -\frac{sR_2C}{1 + sR_1C} = -\frac{R_2}{R_1}$$



AC_OZ4



$$H(s) = \frac{-R_2}{R_1 + sL}$$

$$h(t) = -\frac{R_2}{L} e^{-\frac{R_1}{L} t} \mathbb{1}(t)$$

$$W(s) = \frac{-R_2 L}{s(R_1 + sL)} = \frac{A}{s} + \frac{B}{R_1 + sL} = -\frac{R_2}{R_1} \frac{1}{s} + \frac{R_2}{R_1} \frac{1}{\frac{R_1}{L} + s}$$

$$W(s) = \frac{-R_2 L}{s(R_1 + sL)} = \frac{A}{s} + \frac{B}{R_1 + sL}$$

$$W(s) = -\frac{R_2}{R_1} \frac{1}{s} + \frac{R_2}{R_1} \frac{1}{\frac{R_1}{L} + s}$$

$$A = -\frac{R_2}{R_1}$$

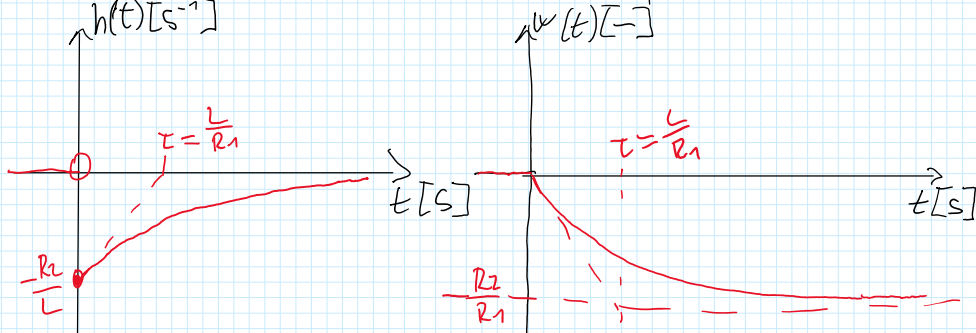
$$B = -AL = -\frac{R_2 L}{R_1}$$

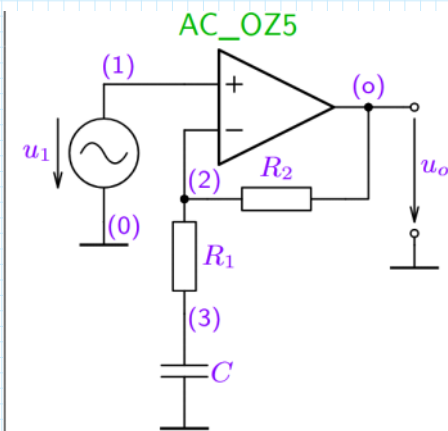
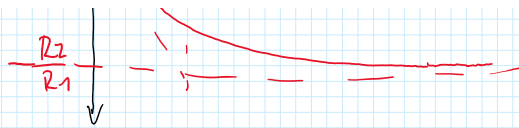
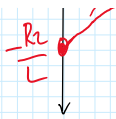
$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} -\frac{sR_2}{R_1 + sL} = -\frac{R_2}{L}$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} -\frac{sR_2}{R_1 + sL} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} -\frac{R_2}{R_1 + sL} = 0$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} -\frac{R_2}{R_1 + sL} = -\frac{R_2}{R_1}$$





$$H(s) = 1 + \frac{R_2}{R_1 + \frac{1}{sC}} = 1 + \frac{sR_2C}{1 + sR_1C} = 1 + \frac{R_2}{R_1} - \frac{R_2}{R_1} \frac{1}{1 + sR_1C}$$

$$h(t) = \delta(t) \left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1C} e^{-\frac{t}{R_1C}} \mathbb{1}(t)$$

$$w(s) = \frac{1}{s} + \frac{R_2C}{1 + sR_1C}$$

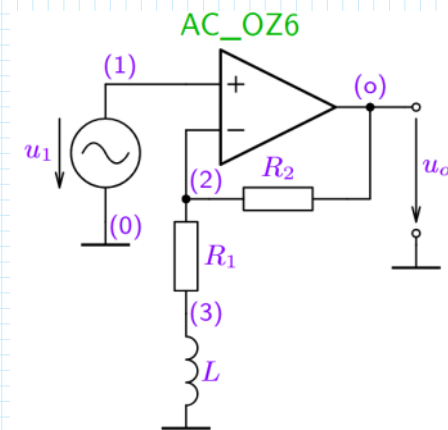
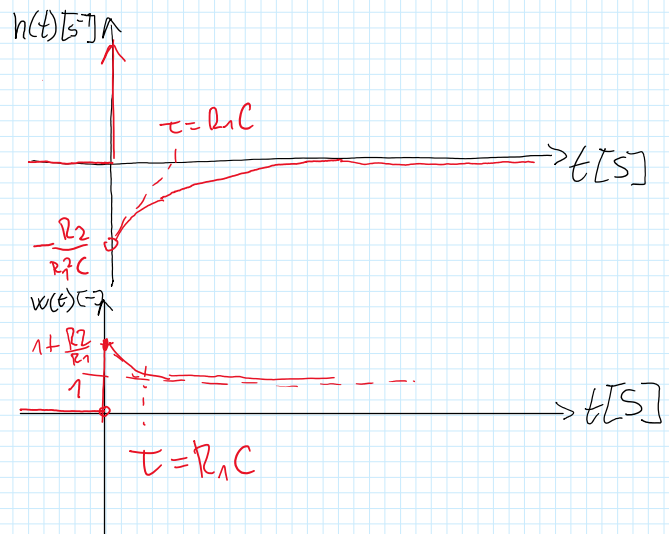
$$w(t) = \mathbb{1}(t) + \frac{R_2}{R_1} e^{-\frac{1}{R_1C}t} \mathbb{1}(t)$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} s + \frac{s^2 R_2 C}{1 + sR_1C} = \infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} s + \frac{s^2 R_2 C}{1 + sR_1C} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} 1 + \frac{sR_2C}{1 + sR_1C} = \frac{R_2}{R_1 + R_2}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} 1 + \frac{sR_2C}{1 + sR_1C} = 1$$



$$H(s) = 1 + \frac{R_2}{R_1 + sL}$$

$$h(t) = \delta(t) + \frac{R_2}{L} e^{-\frac{R_1}{L}t} \mathbb{1}(t)$$

$$w(s) = \frac{1}{s} + \frac{1}{s} \frac{R_2}{R_1 + sL} = \frac{1}{s} + \frac{A}{s} + \frac{B}{R_1 + sL} = \frac{1}{s} \left(1 + \frac{R_2}{R_1}\right) - \frac{R_2}{R_1} \frac{1}{\frac{R_1}{L} + s}$$

$$w(t) = \left(1 + \frac{R_2}{R_1}\right) \mathbb{1}(t) - \frac{R_2}{R_1} e^{-\frac{R_1}{L}t} \mathbb{1}(t)$$

$$R_2 = A(R_1 + sL) + Bs$$

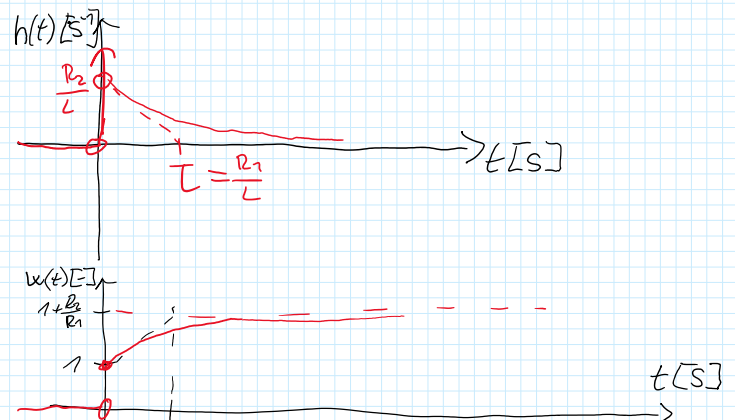
$$\frac{R_2}{R_1} = A \quad B = -AL = -\frac{R_2}{R_1}L$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} s + \frac{sR_2}{R_1 + sL} = \infty$$

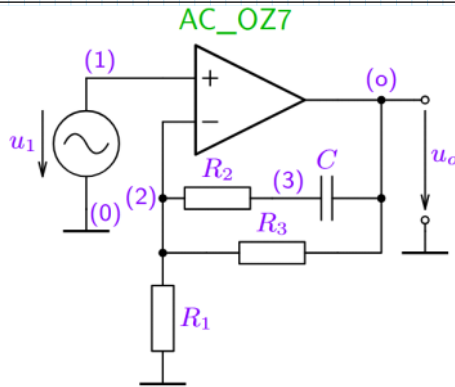
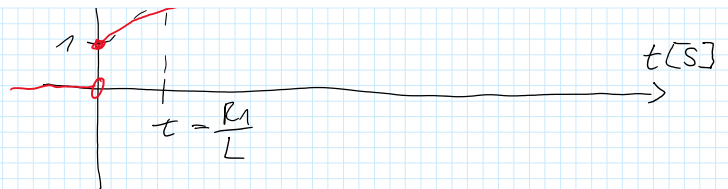
$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} s + \frac{sR_2}{R_1 + sL} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} 1 + \frac{R_2}{R_1 + sL} = 1$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} 1 + \frac{R_2}{R_1 + sL} = 1 + \frac{R_2}{R_1}$$



$$w(t \rightarrow \infty) = s \rightarrow 0 \quad H(s) = s \rightarrow 0 \quad 1 + \frac{R_3}{R_1 + sL} = 1 + \frac{R_3}{R_1}$$



$$H(s) = 1 + \frac{\frac{R_3 \cdot (R_2 + \frac{1}{sC})}{R_2 + R_2 + \frac{1}{sC}}}{R_1} = 1 + \frac{R_3}{R_1} \frac{1 + sR_2C}{1 + s(R_2 + R_3)C} =$$

$$= 1 + \frac{R_3}{R_1} \left(\frac{R_2}{R_2 + R_3} + \frac{\frac{R_3}{R_2 + R_3}}{1 + s(R_2 + R_3)C} \right)$$

$$h(t) = \delta(t) + \frac{R_3}{R_1} \frac{R_2}{R_2 + R_3} \delta(t) + \frac{1}{R_1} \frac{R_3^2}{(R_2 + R_3)C} e^{-\frac{t}{(R_2 + R_3)C}} 1(t)$$

$$W(s) = \frac{H(s)}{s} = \frac{1}{s} + \frac{1}{s} \frac{R_3}{R_1} \frac{1 + sR_2C}{1 + s(R_2 + R_3)C} = \frac{1}{s} + \frac{A}{s} + \frac{B}{1 + s(R_2 + R_3)C}$$

$$W(s) = \frac{1}{s} + \frac{R_3}{R_1} \frac{1}{s} - \frac{R_3^2}{R_1} \frac{1}{1 + s(R_2 + R_3)C}$$

$$W(s) = \frac{1}{s} \left(1 + \frac{R_3}{R_1} \right) - \frac{R_3^2}{R_1(R_2 + R_3)} \frac{1}{\frac{1}{(R_2 + R_3)C} + s}$$

$$w(t) = \left(1 + \frac{R_3}{R_1} \right) 1(t) - \frac{R_3^2}{R_1(R_2 + R_3)} e^{-\frac{t}{(R_2 + R_3)C}} 1(t)$$

$$h(t) = \left(1 + \frac{R_3}{R_1} \right) \delta(t) - \frac{R_3^2}{R_1(R_2 + R_3)} \delta(t) - \dots e^{-\dots}$$

$$= \delta(t) + \frac{R_2 R_3 + R_3^2 - R_3^2}{R_1(R_2 + R_3)} \delta(t) \checkmark + \dots e^{-\dots}$$

$$\frac{R_3}{R_1} (1 + sR_2C) = A(1 + s(R_2 + R_3)C) + Bs$$

$$\frac{R_3}{R_1} = A$$

$$\frac{R_3}{R_1} R_2 C = \frac{R_3}{R_1} (R_2 + R_3) C + B$$

$$B = \frac{R_3}{R_1} (R_2 C - R_2 C - R_3 C)$$

$$B = -\frac{R_3^2}{R_1} C$$

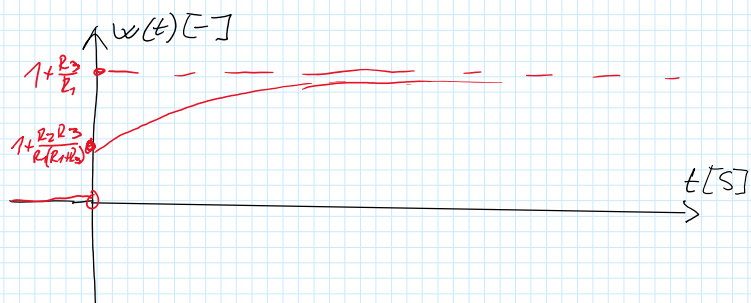
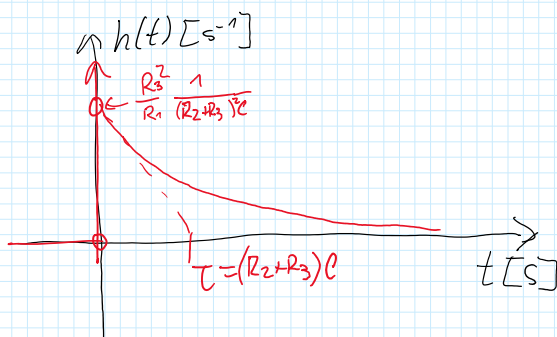
$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} s + \frac{R_3}{R_1} \frac{s + s^2 R_2 C}{1 + s(R_2 + R_3)C} = \infty$$

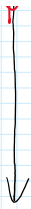
$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \lim_{s \rightarrow 0} s + \frac{R_3}{R_1} \frac{s + s^2 R_2 C}{1 + s(R_2 + R_3)C} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} 1 + \frac{R_3}{R_1} \frac{1 + sR_2C}{1 + s(R_2 + R_3)C} = \lim_{s \rightarrow \infty} \frac{R_1 + sR_1(R_2 + R_3)C + R_3 + sR_2 R_3 C}{R_1 + sR_1(R_2 + R_3)C} =$$

$$= \lim_{s \rightarrow \infty} \frac{R_1(R_2 + R_3)C + R_2 R_3 C}{R_1(R_2 + R_3)C} = 1 + \frac{R_2 R_3}{R_1(R_2 + R_3)}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} 1 + \frac{R_3}{R_1} \frac{1 + sR_2C}{1 + s(R_2 + R_3)C} = 1 + \frac{R_3}{R_1}$$





$$\tau = (R_2 + R_3)C$$

$t [s]$

